

Asexual Reproduction

SCIENTIFIC INVESTIGATION



GROWTH OF LEMONGRASS IN LOAM VS. CLAY SOIL

I. AIM/PROBLEM

- Which soil type produces faster and healthier lemongrass growth over a period of 3 weeks?

hypothesis

- If lemongrass are grown in a loam soil, then it will grow taller and healthier.

II. EXPERIMENT

Materials

- 2 identical cups with drainage holes
- 2 healthy lemongrass cuttings
- 200g of loam soil
- 200g of clay soil
- meter stick
- 100 ml of water

Week	Day/s	Materials Used
Week 1	Day 1	2 small plastic cups, 200g garden soil, 200g clay soil, 2 lemongrass cuttings, 100ml
	Day 2–7	100 mL water (every other day)
Week 2	Day 8–14	100mL water (every other day),
Week 3	Day 15–21	100 mL water (every other day)

Procedure

- Get two cups and put the loam soil in one cup and clay soil in other cup.
- Plant one lemongrass cutting in each pot.
- Water both pots equally every other day.
- Place them in the same spot with sunlight.
- Observe their growth for four weeks.
- Measure and record their height each week.
- Compare which one grows better.

DATA AND OBSERVATION

Data:

TABLE 1

NAME OF THE PLANT (scientific name)	NUMBER OF INDIVIDUALS INVOLVED	TYPE OF ASEXUAL REPRODUCTION	PLANT PART USED IN REPRODUCTION
Lemongrass (<i>Cymbopogon citratus</i>)	2	vegetative reproduction	stem or stalk

Observation:

TABLE 2

DAY		Observation		
7		the lemongrasses are not growing yet		
14		The lemongrass in the loam soil are more longer and higher than clay soil		
21		The lemongrass in clay soil are bigger, longer and higher than the one in loam soil		
Type of Soil	Initial Height (cm)	Week 1 cm	Week 2	Week 3 cm
Garden Soil	10	8	16	12
Clay Soil	10	12	15	24

Guide Questions:

1.What is the independent variable in your experiment

- Type of soil used (loam soil and clay soil)

2.In which set up did your plant grow longer/better?

- The plant grew better in clay soil.

3.If you will repeat the experiment, what changes will you make to grow the plant better/longer?

- If I will repeat the experiment, I will add more water and place the plant in a sunnier area and plant it in a clay soil so it can grow better and longer.

Conclusion

- Therefore, the tanglad grew better in clay soil because it was able to hold more moisture, which helped the plant stay healthy and grow longer.

LEMONGRASS FROM DAY 1-21

Day 1



Day 2



Day 3



Day 4



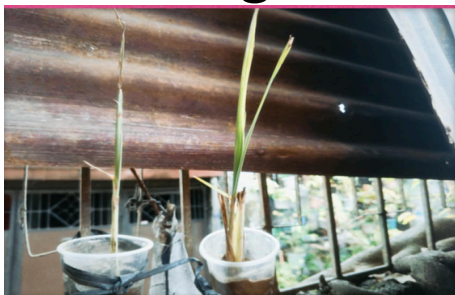
Day 5



Day 6



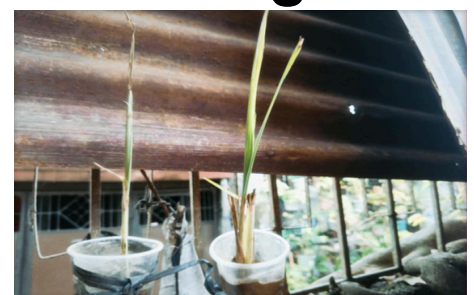
Day 7



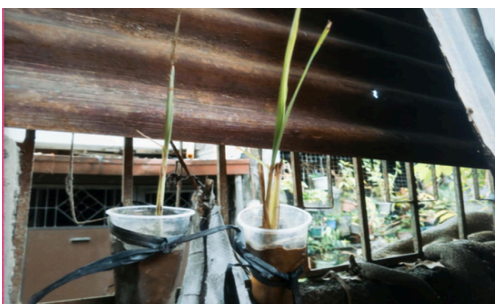
Day 8



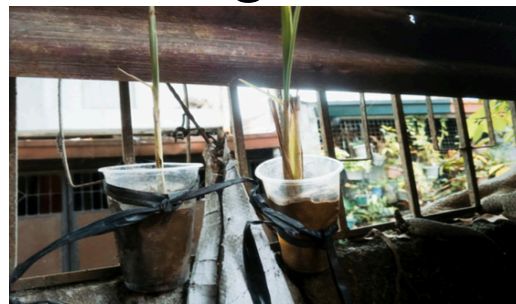
Day 9



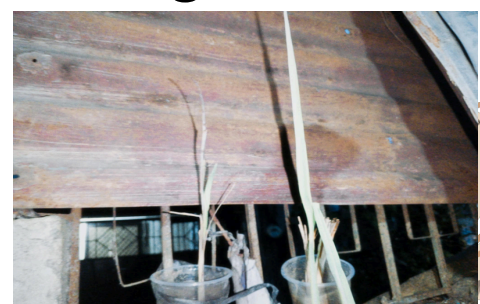
Day 10



Day 11



Day 12-13



Day 13-16



Day 16-18



Day 18-20



Day 21



INDEPENDENT VARIABLE

- Type of soil (garden soil and clay soil)

DEPENDENT VARIABLE

- Growth of tanglad (height and health of the plant)

CONTROLLED VARIABLE

- Amount of water, amount of sunlight, size of container, and number of tanglad plants used

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